Manual

Failure Mode Analysis

PDV-FSA 8.1

Version 1.0.23049

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Overview

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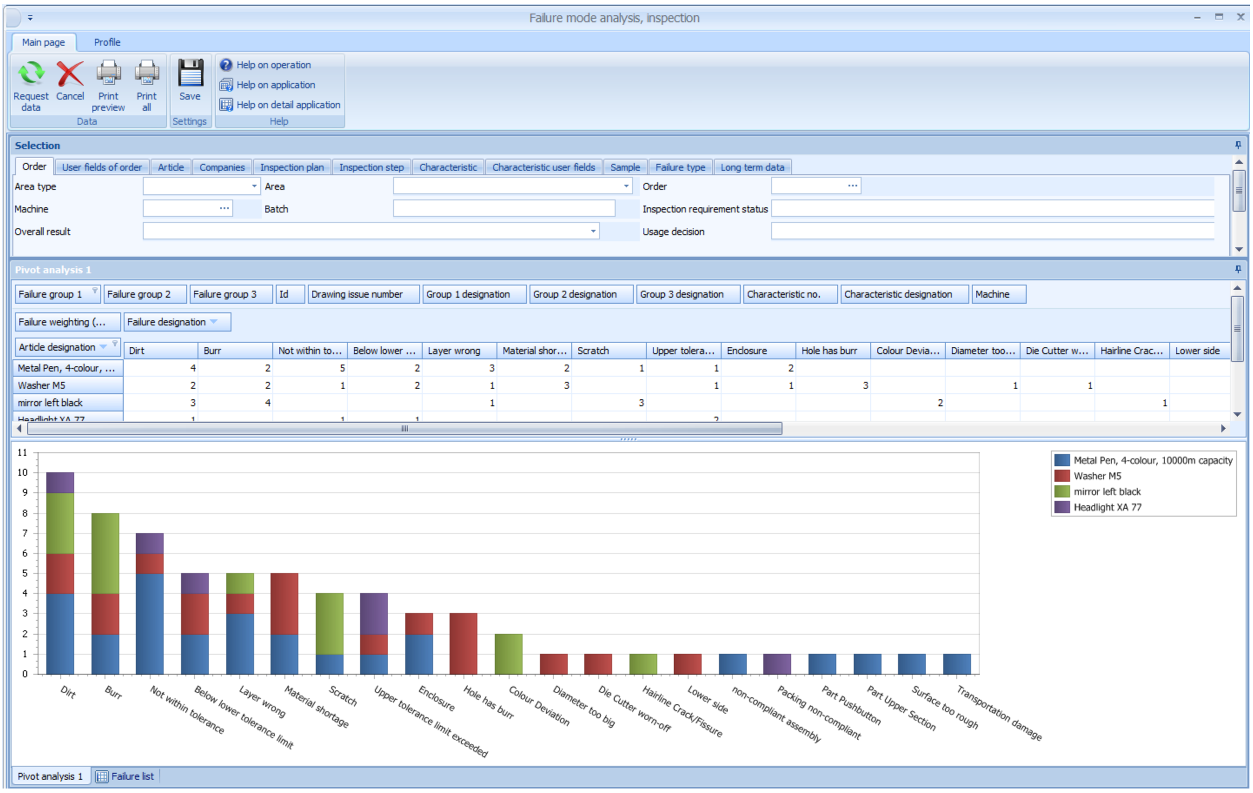


Purpose

Use the *failure mode analysis* to evaluate failures of the following types. These failures are collected during the inspection process or generated automatically.

- Failure type (FT)
- Failure location (FL) and
- Failure cause (FC).

Use the available pivot functions to evaluate the data. Using these functions, you can display the distribution of defect types (frequency) for each article/item over a specified period of time. This analysis helps you to specify core areas that might require action to optimize the quality in the production process.

Integration

The *failure mode analysis* evaluates failures of inspection requirements from the area types:

- Goods receipt
- Initial sample inspection
- Production
- Goods issue

Consequently, the failure mode analysis is a general analysis tool.

Requirements

There are no special requirements. You only have to record the failures. To do so, you have to define the failures beforehand in the master data of Quality Management.

Selection criteria

The following list shows some of the available selection criteria. Self-explanatory filter options are not listed.

Order tab

Area type

Selection list of area types
- in-production inspection
- goods receipt inspection
- goods issue inspection
- initial sample inspection

The list of area types depends on the licenses in use. The list can be restricted.

Area

Selection list of the configured areas matching the previously filtered area type. By default, the following areas are available:
- in-production inspection: production
- goods receipt inspection: goods receipt
- goods issue inspection: goods issue
- initial sample inspection: initial sample

You can add further areas through customizing.

Inspection requirement status

Selection list including multiple selection options for all inspection requirement statuses.

Overall result

Selection list for the results of inspection requirements (pass, fail, conditionally pass).

Usage decision

Selection list of usage decisions (e.g. release, special permit, rework, reject, sort) that are available, once an inspection requirement has been completed.

***Inspection point identification* tab**

If these fields are not available, you require a new program version of this application.

Workplace

Workplace number

Partial batch

Number of partial batch

ERP batch

ERP batch number

Field 1…3

Additional fields for inspection points

***Additional fields for inspection points* tab**

If these fields are not available, you require a new program version of this application.

Field 4…8

Additional fields for inspection points

If required, the fields *Field 1* to *Field 8* are enabled as part of an MPDV customization. The field labels are assigned individually. As these field names are customizable, they are only entitled *Field 1* to *Field 8* in this document.
Field 1 includes, for example, the tool if cavity-related data collection is enabled.

***Sample* tab**

Sample machine

Selection list of machines/workplaces
The machine assigned to the first measured value is displayed here, provided that different machines are assigned to the measured values of the same sample.
The application assigns the machine where the failure was recorded to the failure. If the failure was recorded at an inspection station, the system assigns the machine number of the inspection station to the failure.

Machine group

Selection list of machine groups
The application shows the failures that are assigned to a machine pertaining to the selected machine group.

Invalid

The application shows the failures whose superordinate sample is valid or invalid. If necessary, you can also filter the valid **and** invalid failures.

***Failure* tab**

Failure time

You can filter data by the time the failure was recorded.
If you activate the license FEP-AQF, the *failures* tab does not show the fields you can use to filter by the failure time. But you can use filter fields to restrict the displayed failures by the shifts and the shift date and time.

Date, shift, time

These fields are only available if you activate the license "FEP-AQF".

If you select the radio button *shift* and enter a *from/to* date, the application filters the data by the shift date and the selected shift number. The date filter field is automatically set to the shift date. The shift date corresponds to the start date of the shift of the corresponding machine. The system identifies the machine where the corresponding order is logged on.
If you select the radio button *time,* the application filters the data by the date and time when the failure analysis entry is recorded.

If you filter data by the shift and optionally enter the shift date, the application only filters those failures where the superordinate inspection point was generated after the license FEP-AQF has been activated. All inspection points recorded before the activation of the license neither show a shift number nor a shift date. If you want to filter failures by the shift, you have to collect data in relation to inspection points.

Toolbar

There are no other special function buttons in addition to the standard functions/features.

Graphic failure analysis detail applications

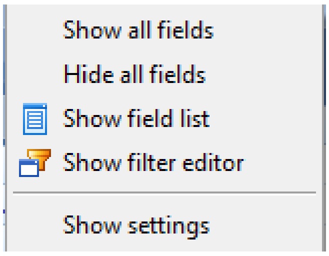
Data is presented in a pivot table including bar charts. Different application functions are available to display data. The filtered failures represent the data basis for evaluations.

This document does not describe the general pivot functions in more detail. The paragraphs that follow only describe the basic functions of this evaluation/report.

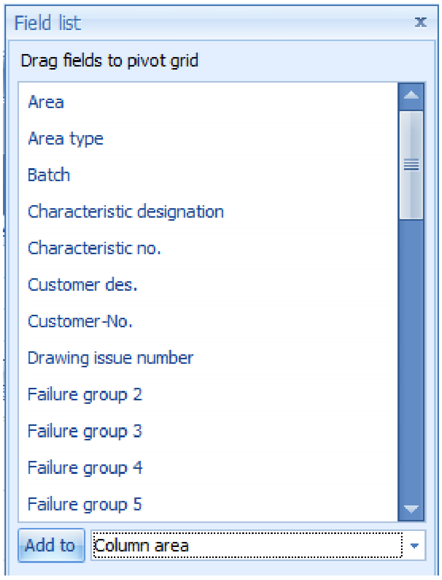
Pivot evaluations/reports provide the following benefits:

- You can quickly summarize and present large amounts of data.
- You can exchange rows and columns to have the source data summarized differently.
- You can filter data easily using drag and drop and additional detail filters.
- You can use the interactive way of presentation to summarize and analyze data in different formats and using different calculation methods.

Right-click to open the below context menu:

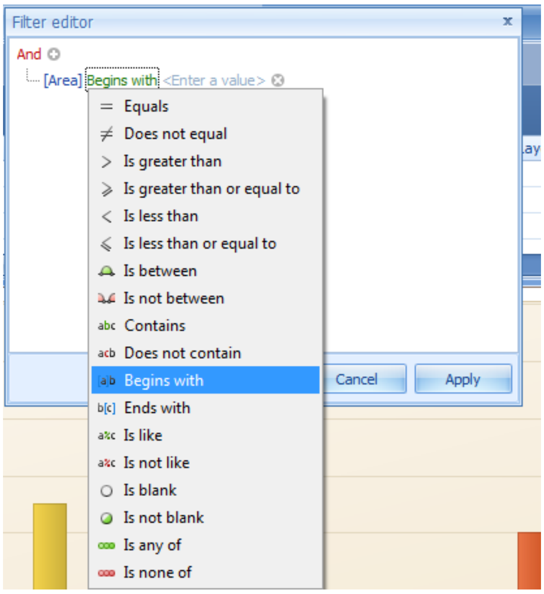


You can use the function *Show field list* to select the fields you want to use for the pivot analysis. The below figure shows what a field list could look like.

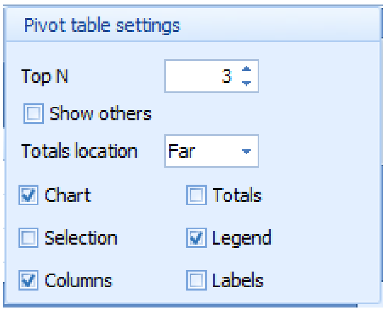


You can drag the required fields and drop them in the evaluation area (drag and drop).

In addition to the selection criteria, you can use the function *Show filter editor* to further narrow down the data displayed.

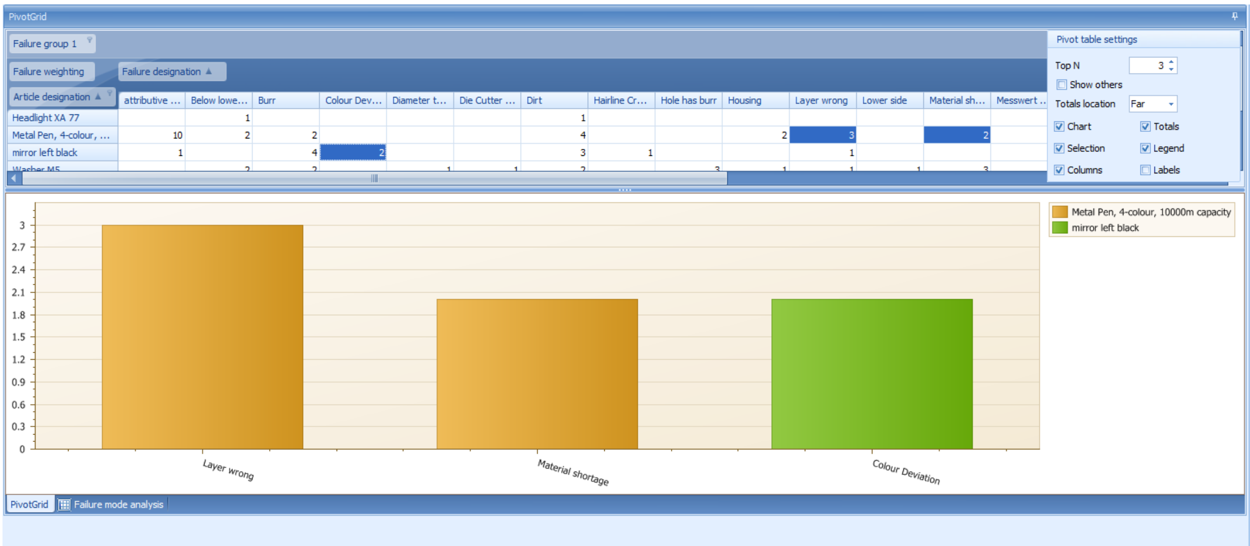


If you click *Show settings* in the context menu, the below dialog opens:



Check the *Selection* option and select a specific area to specify the contents of the tabular display. In this case, the graphic representation is based on the selected cells. Check the *Label* option to display the total number for each bar.

The below figure illustrates these functions.



Check the *Totals* option to display the row *Overall result* in the bar chart. If you check the *Selection* option, the pivot report is only based on the data of the selected cells. The application also uses the selected cells to identify the overall result.
Check/uncheck the *Columns* option to switch between the graphic presentation of the corresponding number of columns and rows.

Failure list detail applications

The failure list shows the failures including referenced data matching the entered selection criteria. The referenced data usually corresponds to the field list for the pivot analysis.